Tomer Gafni

Postdoctoral Researcher
Department of Industrial Engineering and Operations Research
Columbia University

CURRICULUM VITAE
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Research

My research lies at the intersection of information theory, statistics, and machine learning. I develop theory and algorithms for sequential decision-making under uncertainty and non-stationarity, including multi-armed bandits, active hypothesis testing, and change detection. I have also worked on efficient quantization for deep neural networks.

Appointments and Experience

2025–present **Postdoctoral Researcher, Columbia University**
Department of Industrial Engineering and Operations Research.

2024–2025 **Algorithm Developer, Intel**
Quantization algorithms for large language models.

2023–2024 **AI Researcher, Elbit Systems**
C4I & Cyber. Machine learning for communication.

Education

2019–2024 **Ph.D., Electrical and Computer Engineering**
Ben-Gurion University of the Negev
Dissertation: *Online Learning for Efficient Communication*
Advisor: Prof. Kobi Cohen

2018–2019 **M.Sc., Electrical and Computer Engineering**
Ben-Gurion University of the Negev, Meitar accelerated program
Thesis: *Learning in Restless Multi-Armed Bandits for Cognitive Radio Networks*

2014–2018 **B.Sc., Electrical and Computer Engineering**
Ben-Gurion University of the Negev; magna cum laude
Tracks: Signal Processing and Communication

Selected Honors and Awards

2025 VATAT Fellowship for Outstanding Postdoctoral Researchers in Data Science, Council for Higher Education, Israel.

2023 Dean Excellence Award for 2022 Ph.D. achievements.

2022 Kaufmann Award for the Highest Student Achievement in Electrical Engineering, Ben-Gurion University.

2022 Darom-Besor Fellowship for outstanding students in the combined Ph.D. program.

2020 Teaching Excellence Award, Introduction to Modern Communication.

2019 Darom-Lachish Fellowship for outstanding students in the combined Ph.D. program.

Publications

Journals and Selected Conferences

J1. **Tomer Gafni**, Garud Iyengar, and Assaf Zeevi. "The Cost of Learning under Multiple Change Points." *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, Seoul, Republic of Korea, July 2026.

J2. Liad Lea Didi, **Tomer Gafni**, and Kobi Cohen. "Asymptotically Optimal Search for a Change Point Anomaly under a Composite Hypothesis Model." *IEEE Transactions on Signal Processing*, vol. 74, pp. 1880–1896, 2026.

J3. Yaniv Cohen, **Tomer Gafni**, Ronen Greenberg, and Kobi Cohen. "SINR-Aware Deep Reinforcement Learning for Distributed Dynamic Channel Allocation in Cognitive Interference Networks." *IEEE Transactions on Wireless Communications*, vol. 24, no. 1, pp. 228–243, January 2025.

J4. **Tomer Gafni**, Kobi Cohen, and Yonina C. Eldar. "Federated Learning from Heterogeneous Data via Controlled Air Aggregation with Bayesian Estimation." *IEEE Transactions on Signal Processing*, vol. 72, pp. 1928–1943, 2024.

J5. **Tomer Gafni**, Benjamin Wolff, Guy Revach, Nir Shlezinger, and Kobi Cohen. "Anomaly Search over Discrete Composite Hypotheses in Hierarchical Statistical Models." *IEEE Transactions on Signal Processing*, vol. 71, pp. 202–217, 2023.
Tomer Gafni and Benjamin Wolff contributed equally.

J6. **Tomer Gafni**, Michal Yemini, and Kobi Cohen. "Learning in Restless Bandits under Exogenous Global Markov Process." *IEEE Transactions on Signal Processing*, vol. 70, pp. 5679–5693, 2022.

J7. **Tomer Gafni**, Nir Shlezinger, Kobi Cohen, Yonina C. Eldar, and H. Vincent Poor. "Federated Learning: A Signal Processing Perspective." *IEEE Signal Processing Magazine*, vol. 39, no. 3, pp. 14–41, May 2022.
Highlighted in the IEEE Xplore Popular Articles list.

J8. **Tomer Gafni** and Kobi Cohen. "Distributed Learning over Markovian Fading Channels for Stable Spectrum Access." *IEEE Access*, vol. 10, pp. 46652–46669, 2022.

J9. **Tomer Gafni** and Kobi Cohen. "Learning in Restless Multiarmed Bandits via Adaptive Arm Sequencing Rules." *IEEE Transactions on Automatic Control*, vol. 66, no. 10, pp. 5029–5036, October 2021.

J10. **Tomer Gafni**, Kobi Cohen, and Qing Zhao. "Searching for Unknown Anomalies in Hierarchical Data Streams." *IEEE Signal Processing Letters*, vol. 28, pp. 1774–1778, 2021.

J11. Bar Hemo, **Tomer Gafni**, Kobi Cohen, and Qing Zhao. "Searching for Anomalies over Composite Hypotheses." *IEEE Transactions on Signal Processing*, vol. 68, pp. 1181–1196, February 2020.
Bar Hemo and Tomer Gafni contributed equally.

Publications (continued)

Preprint

P1. **Tomer Gafni**, Garud Iyengar, and Assaf Zeevi. "Data Reuse in Non-Stationary Learning." arXiv:2610.10340, October 2026.

Conference Proceedings

C1. **Tomer Gafni**, Asaf Karnieli, and Yair Hanani. "Dual Precision Quantization for Efficient and Accurate Deep Neural Networks Inference." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 3284–3294, June 2025.

C2. Liad Lea Didi, **Tomer Gafni**, and Kobi Cohen. "Active Change Point Anomaly Detection over Composite Hypotheses." *Proceedings of the 60th Annual Allerton Conference on Communication, Control, and Computing*, pp. 1–6, September 2024.

C3. **Tomer Gafni**, Benjamin Wolff, Guy Revach, Nir Shlezinger, and Kobi Cohen. "Anomaly Search over Discrete Composite Hypotheses in Hierarchical Statistical Models." *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2023.

C4. **Tomer Gafni**, Kobi Cohen, and Yonina C. Eldar. "CoBAAF: Controlled Bayesian Air Aggregation Federated Learning from Heterogeneous Data." *Proceedings of the 58th Annual Allerton Conference on Communication, Control, and Computing*, pp. 1–4, September 2022.

C5. Benjamin Wolff, **Tomer Gafni**, Guy Revach, Nir Shlezinger, and Kobi Cohen. "Composite Anomaly Detection via Hierarchical Dynamic Search." *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, pp. 2421–2426, July 2022.

C6. **Tomer Gafni**, Michal Yemini, and Kobi Cohen. "Restless Multi-Armed Bandits under Exogenous Global Markov Process." *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.

C7. **Tomer Gafni** and Kobi Cohen. "A Distributed Stable Strategy Learning Algorithm for Multiuser Dynamic Spectrum Access." *Proceedings of the 57th Annual Allerton Conference on Communication, Control, and Computing*, September 2019.

C8. **Tomer Gafni** and Kobi Cohen. "Learning in Restless Multi-Armed Bandits Using Adaptive Arm Sequencing Rules." *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, pp. 1206–1210, June 2018.

Teaching

Teaching Assistant, Ben-Gurion University 2018–2023

Introduction to Linear Systems (Spring 2018–2023); Introduction to Stochastic Processes (Fall 2019–2023); Introduction to Modern Communication (Fall and Spring 2018–2021).